

Purposefully Nonstandard Cybernetics an alternative to purpose driven control

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okay the next talk is going to be by Bradley Alisia and colleagues purposefully non-standard cybernetics an alternative to purpose-driven control this is going to be a 1our pre-recorded talk so thank you enjoy hello my name is Dr Bradley Al and on behalf of Jesse parent Morgan huff and Anda Nelson I'm we're from the orthogon research and education laboratory and the cybernetics interest group and I'm going to talk to you today about per purposefully non-standard cybernetics an alternative to purpose-driven control so we start with a question what is the definition of behavior and this is varied throughout history according to the behaviorist Behavior was identified as a reflex elicited by stimulus associations the consequence of reinforcement and the current motivational state if we go forward to the cognitive ists which is more or less the modern view they would say that behavior is a set of internal mental states that are responsible for observable behaviors with gold directed end States now it's of note that both behaviorist and cognitivist Views involve feedback this leads to learning a notion of what constitutes a state and predictive capacity so we start with this paper Rosen blue th all in 1943 Behavior purpose and teeology published in the philosophy of science and it's a formative paper in the field of cybernetics so in that paper they have this typology of Behavioral classifications so we're going to return to this typology throughout the talk and we're going to think about how we can reorganize this typology but it basically defines Behavior as a set of nested things the red text Non classes or they're basically all other things other than the uh black thing that we'll be talking about at each rung of this typology so behavior is active it's purposeful it involves feedback which implies some sort of chology or end goal it's also predictive which involves extrapolation and involves an order of prediction so you might have first order second order or third order prediction so to understand what teeology is and where it comes from we start at the Middle Ages where teeology was the end State or the end goal of some behavioral process was thought of as a Divine causation we then move forward to teeology As Natural cause a and then it branches off into two directions and the first was the

vitalism of bergson and evolution by natural selection made famous by Darwin leading from Darwin's work we had arar who talked about tonomy which is purpose versus history and we'll talk a little bit about that later in the talk kind of discontinuous to this are two different streams so the first at the top is where behavior is viewed as habit that's by William James and Thorndike that leads us to the an molec K antology of Francis CRI and feedback is complexity which is summarized by control theory and dynamical systems at the bottom is our target paper that we just talked about these are the views of roseen blue thin Wier and this is feedback relative to a goal this led us to cognitive science and reinforcement learning at least the modern versions of this where people talk about gold directed behavior and variations on that in molecular biology such as molecular vitalism and the tame and U methodology IES of leban janca and Ginsburg so we'll talk about a couple papers here and frame some of this in context so if we think about Behavior it's it's essentially a form of functionalism in the first paper here Rody and Germain they argue that functionalism is a pre-scientific relic so they actually argue that the whole idea of functionalism is something that comes from this era of divine causality biology is not designed so why should we expect a function and this is something that of course maybe is assumed about function that it comes from some sort of design or comes from some sort of divine intervention on the other hand toonomic systems are defined by spatial and temporal information and constant iterations Over States so we do have these toomic systems but include modern components only select behaviors can be defined as selection of a goal or derived from signals from the goal so that means that we can't necessarily paint Behavior with a broad brush of teeology only some behaviors may be theological and others maybe not not all behaviors and this includes biological and machine behaviors are goal selecting and while goal directed behaviors seem to be purposeful and it may be true that some of these behaviors that are known as gold directed are purposeful are there Developmental and evolutionary antecedence so are those Place those uh ancestral behaviors or those basil behaviors or behaviors that occur earlier in development are those also purposeful and the answer may be no so Rosen Lu at all Define behavior as any change in an agent with respect to its surroundings this is much broader than the Neuroscience psychological definition and it sets up a contrast between all behaviors and a subset of purposeful behaviors so if a behavior is put under some sort of selection or if a behavior is selection of something in the environment this might lead us to a subset of voluntary behaviors which then some of those may be purpose ful and so purpose is actually purposeful behaviors is actually a subset of all behaviors to demonstrate this we see that clocks both mechanical and

biological are not purposeful as there is no final aspirational state mechanical clocks can keep time and you can set an alarm on them but that alarm is imposed by the outside and it's not a property of that behavior and of itself further more biological clock runs via physiological processes like a circadian rhythm and it can be entrained by external white but that again is not a final aspirational State more generally teleology requires a signal from a goal so it's some sort of signal from a goal that's specific to that behavior and purposes progress to that goal so again with our uh example of circadian rhythms it's entrained by light but that light itself is not a goal it's just something that's out in the environment and the system entrains itself to it we can also see this with brains in that brains produce behavioral outputs but the exact mechanism is hard to pinpoint and this is the Rost and Lal article that we saw earlier and this asks the question signal from what to goal so there's a signal coming from something to a goal but it's not clear what that is this is something one of our co-authors Jesse parent works on this is a frontier map for teleology and related complexity Theory Concepts so the data from this is drawn from the complex world book by crick hour this is published in 2024 and it kind of goes over three different concepts that are interrelated so we have teleology organized complexity and emergence and so you can see that our paper Rosen Bluth 1943 is at the top and this is sort of the first paper that talks about teleology in sort of a scientific and in a a manner of complexity and so this is the first paper that talks about teleology is a property of complexity and of complex systems so a lot of papers were influenced by Rosen Bluth this uh teleology was inherited by Weiner's book cybernetics and control in 194 two other papers adaptive systems in uh by John Holland in 1962 and organizations and complexity also in 1962 the good regulator of course in 1970 inherited this concept but it also inherited another concept from organizations and complexity which is organized complexity the good regulator then passes down this concept of teleology to the book order and chaos in 1993 and there's a direct link between Rosen Bluth 1943 and the tame concept of Mike 11 in 2022 now we going back to the concept of organized complexity this comes from organizations and complexity the good regulator in 1970 and the autopoesis paper in 1974 both share this intellectual lineage the autopoesis paper lends that organized complexity concept to this paper on deiv in 1982 get to this book order and chaos in 1993 and then to the tame Concept in 2020 2 finally we have emergence which starts with the paper by Anderson war is different in 1972 inherited by the autopoesis paper 1974 in turn inherited by the dissipativity paper 1982 in turn inherited by Order and Chaos 1993 so you can see that we have not just teleology as a concept in which we've kind of talked about already but also a number of other related Concepts this is 11 uh

this is the tame paper and then Janca Ginsburg this is the U paper and so Lan proposes the tame or technological approach to mind everyone Lev's principle here is this cognition all the way down we can see cognition in terms of Minds we can see cognition in terms of molecular phenomena we can see cognition everywhere in between and so behaviors all those behaviors are cognitive to some extent and as a result we get these diverse Minds that have these different goal directed competencies that allow those systems to achieve embodied agents so in Levan's approach you know kology is a very important concept not just for cognitive type behaviors but behaviors that that go all the way down to the molecular scale and maybe even down to the quantum scale second theology is a means to explain why agents develop towards goals and so you know we can look at uh agents such as artificial agents simple organisms like bacteria then we can see that like teeology is a way to explain learn how to move towards goals the next paper is this U paper by Janca at all and in that paper they propose a concept called unlimited associative learning and this concept is a minimal marker for Consciousness so in this paper they also Champion teeology is an intrinsic reinforcement system you have this inter internally regulated model in service of attaining external goal andology plays a role in attaining goals so again in both cases there's a strong tendency to talk about goals the strong tendency towards to but we might ask the question is behavior at these different scales and any sort of behavior that sort of motivated towards some sort of function ta logical and purposeful or is it simply a product of the history of the system so you know we could talk about teeology and purpose and we've kind of set this up that you know this is something that has kind of come out of the natural study of behavior but sometimes you know we don't really fully you know sometimes we ascribe heliology and purpose to things that maybe are just simply a product of the history of a system first example I want to give is the basov jabotinsky reaction or the bz reaction and at the lower left you can see a image of the bz reaction and this is a deterministic internal model that we can model with differential equations and we can solve those differential equations and get interesting solutions that produces a cave chaotic behavioral output or the Dynamics of the system which are sort of the unfolding history of the system so if we take an example of some sort of chemical soup we can achieve bot Jaa tinsky reactions or bz reactions by setting up this sort of set of differential equations these ordinary differential equations setting up an initial condition and we get these spiral waves and scr wings that result and so if we have slight variations in initial condition we end up with very different systems very different types of patterns in these um in these systems and we can describe this with ODS and one might say okay there's a purpose to this but the purpose isn't

necessarily clear when you run this simulation again and again with different types of initial conditions there's a chaotic Behavior at a variational dynamic dynamical output so this is an example of the tip trajectories of BZ waves this is from a paper I Bron an angle in 1993 and it shows how there's a lot of variation in the pattern but there's a sort of um you know constant aspect of the pattern in terms of the tip trajectories so one might say okay the purpose is to produce these tip TR trajectories instead of the actual pattern and the interactions which may be true but then of course you know that's something something that I want you to think about as we go through the rest of this talk we could also plug the bosov jabotinsky reaction into a cybernetic model with higher level feedbacks and so I talk about this in this dimensions of morphogenesis lecture on the dorm YouTube channel and what I'm getting at in that talk is that there different dimensions of morphogenesis there's onedimensional morphogenesis two-dimensional morphogenesis higher dimensional morphogenesis and so you see this kind of output and so I'm asking the question here is this purposeful or is this a product of sort of a a chemical a dynamical chemical system with a history and so we can also look at this as a cybernetic model where we have feedbacks and we have other sorts of complexity and you know we can fit it also into that behavioral uh typology of roseen blue okay another uh another example is this thing called an allostasis machine so an Al estasis machine is is basically the output of an internal model and so an internal model of course as we talked about is something that has some sort of capacity for memory and learning and maybe some sort of output that looks Behavioral or cognitive in some way so what we can do is we can take this internal model we can have an oscillator which is a stochastic process or a deterministic process depending on our oscillator and we can have a memory with capacity so we have this oscillator that produces an output and we have sensory information that can drive the oscillator forward this output creates a trajectory which is an interaction with between noisy perturbations in a historical trajectory that's shown in this red line and the historical trajectory is stored in the internal model was a memory with a capacity so as this red line unfolds there's a memory of it in the internal model and it responds to external perturbations which drive this uh Red Line in different directions and the oscillator tries to recover to the original state and so you get these dynamics that may look purposeful but really have this stochastic and adaptive component we get more into this aspect of dynamical control systems I look at intrinsic motivation in dynamical control systems so this is this tiomkin paper PRX life and so in this paper they talk about how you might model maybe something like an alisasas machine as a set of actions generated without a specific reward signal so in this paper they talk about that as

intrinsic motivation they model this with except States and AST actions so an action maps to a number of states and this results in a trajectory forced by observation stochasticity this is entropy so you produce the trajectory there's an observation that you have and that's the entropy of the system and so you maximize actions that increase susceptibility rather than entropy these are noisy observations that we do in perfect observational States and you can compare par arbitrary length sequences of such sequences in the future this gives your internal model the ability to sort of integrate information and to find sort of solutions to problems the idea is that you have the system that can work towards a goal but it isn't driven by any sort of teleology it's just driven by this memory component and the stochastic component and that's all you need a number of other papers here that I'm going to talk about I'm going to put this in the framework of anticipatory systems from our paper on tonomy and any Tristen so to get to this we need to re kind of reformulate our behavioral typology so remember we talked about behavior is having these different components we have these non- classes so in this case we're going from Behavior to active Behavior a purposeful Behavior but that's contrasted with historical or markovian behavior then we have feedback prediction and Order of prediction and so ERM refined the Rosen blue that all theological category to Parts purposeful agentic and historical marvian so if we go back to the Benny and friston paper they talk about the free energy principle is managing different forms of gold directedness and how you need negative feedback to attain homeostasis there's purpose in in cases where the goal is known which is a marvian system and surprisal was minimized which resolves uncertainty then we can expect purpose purpose isn't really just some sort of magical thing or it's not some sort of thing that requires is a strict goal it's just certain conditions under which our behavioral trajectory exists we can also look at anticipatory systems it's the louy paper and this anticipatory systems come from a generative model with depth so this is where we have temporal aspects of the system that lead us to teleology so it's more that the teleology is the history of the system and this depth comes from an internal model so basically if we go back to the internal model we have some process that generates Behavior we have this memory the memory is basically something that stores everything that the system has experienced and that's how we get to kind of understand this a little bit more we want to distinguish between a marvian system and a non-markovian system so our maravian system is where you have inter arrival times so there's a series of things coming in in time this is exponentially distributed in this diagram we have sensory information coming in we have some sort of encoding this encoding is a binary Channel with noise where we have these lines that represent some sort of sort of sensory input

or perturbation of the system and the inner arrival times the area between these lines so the initial arrival time is obviously not equidistant in this example but it's in between the red and blue lines or the red and red lines and so we can take those in our arrival times and if they're exponentially distributed we have a Markovian system if they're not then we have a non-Markovian system and so we've kind of developed this uh distinction between Markovian and non-Markovian sensory information and how that relates to an information measure in our Gibsonian information work so Gibsonian information is a Markovian but can also be non-Markovian information processing system and the citation is here so going back to that behavioral typology we have active purposeful feedback prediction and Order of prediction feedback is everything that is circular non-transitive this is the present p past and future your behavior is controlled by a margin of error Corrections of agent location the time t_e relative to a goal position or state and so if we think about this in terms of a feed forward process feed forward process is where the past is your input present is your process and the future is your output in feedback however it's a little bit different your input comes in the past and then you have this present State here and then you have feedback which is the past and that loops back to your input which means that your input is now the future and so the future is actually present with past feedback so you see the loop here you have input to the present you have this feedback which is the past and it goes to the Future so now you have this future State before the present State and if that keeps going out you get this all output with alternative Futures so your feedback loop can be positive or negative and in this case it doesn't matter what we're interested in is sort of the order of past present and future so we end up with this input we go to the present we go to the past we go to the future we go back to the present so we're sort of interdigitating different types of causality here instead of just having this past present and future aspect to things we have this sort of convoluted aspect of past present and future if we go back to our historical map we recall that non-olical things can be characterized as feedback as complexity and so there are two papers here that talk about pit control that kind of get us at that idea so pit control is proportional integral derivative control and if we think back to our example of chemotaxis and phototaxis we find that PID control can be applied to chemotaxis and other types of biological control so this is The Balter and Buckley paper they talk about this in terms of a variational free energy minimization given a linear generative model they talk about pit control resembling a reflex arc mechanism so this is the reflex arc we find in sensory motor control and the free energy principle provides a more direct account of control this is where we don't have any hidden states in the process of proprioception so they're

focusing on proprioception and but they also talk about chemotaxis and so they kind of give us this sort of uh variational free energy interpretation of this um in go all which is a more engineering take on this the pi and PD components or the proportional integral and the proportional derivative components and directed Network synchronization are all sort of uh components of this control mechanism in particular the pi and PD components of PID Define present past and future components in an internal model control system we can then also start to think of feedback as complexity in terms of nonlinearity and complexity as well so this paper by Westerberg at all stimulus history not expectation Drive sensory prediction errors in a melean cortex this is from the bio archive so the author has framed this in terms of predictive coding so predictive coding explains cortical responses in two ways a feed forward model where an internal model of expected events occurs or in terms of an internal model of expected events or a feedforward approach and a calculation of prediction error for unexpected stimuli or feedback approach the latter are actually Global oddballs which defy expectations in a local context so in Mouse and macak which are their two model organisms they see two types of signal they see a normal response where the internal model is matched by little stimuli and a global Oddball response which is emerged which emerges as a higher order response to this normal response we also can look at this nonlinearity in terms of Behavioral output in general and then burstiness or nonnormality of so this paper burstiness of human physical activities and their characterization and quantifying postural sway Dynamics using burstiness and interevent time distributions kind of gets at both this aspect of Marian systems and and a behavioral system that exhibits sort of nonlinear complexity both papers focus on the burstiness of postural sway in postural sway we have this Paradigm where someone stands on a plate and that plate is a force plate that measures the center mass of the individual and then we can measure this and we get this nice dynamical trajectory that has different features that we can analyze so it's a behavioral output that exhibits a lot of nonlinear characteristics and dynamical characteristics teuchi and Sano focuses on burstiness and postural sway in terms of adult housework and children's play behaviors they studied physical activities of children between ages two and five and they found that burstiness is not an acquired feature it's something that's innate burstiness is present in healthy healthy individuals not in people with movement disorders or impairments they also looked at inter event time distributions and Associated fluctuations so we talked about inner event times as being key to more Marian and non- Marian systems they found that they're bursty so they're these stochastic signals that have a noise component and sometimes this is in function of someone

trying to maintain their balance and sometimes it's just noise that's generated by the nervous system that's added to the stochastic signal in both cases This Is Not Gold directed it's generated by physiological system it's a very complex behavioral signal and it's not explainable by either a pan or gaussian model which means that it has this very complex set of features so if we do some experimental cybernetics on this we can find that if we have the right internal model we can produce very interesting output signals or behavioral signals some of these T kind of look like they could be chological but they're not this is just simply running some stochastic sign one example is just a pure stochastic signal the other one is having a stochastic signal with feedback that then takes that signal and integrates it at a later date so what we have here are two Fe forward processing one is a spway stochastic signal in Blue on the left and then the other is a sum signal that is STO a stochastic delay of that signal or the signal on the right so on the right we have the blue signal you can kind of see it under the red signal so the red signal is the one that's bursty and it's is basically where at random intervals we sum previous output and we add it to the present output so we end up with these bursts and so the delay which is this uh sum signal that we uh that represent our bursts is the buildup of the input signal it produces these bursts or what we might call avalanches and so we can see that we can produce these nonlinear signals from a simple internal model so this is the classic coopel paper from 1982 it actually has a couple of interesting things to say of cybernetic oriented things so we have this feed forward which is where you have a process that goes from A to B to C to D and we also have these systems that have strong back coupling so we have a process that goes from A to B to C to D then we have these back couplings from D to C C to b b to a and so what's more likely to be the case of like you know actual sort of complex behavior is not the feed forward signal but the strong back coupling this is where we have ubiquitous feedback Str back coupling can be intractable to analysis but produces the most interesting aspect of network interactions and strong back coupling produces things like Collective Behavior emergent properties stochastic outputs and nonlinear outputs or Avalanches which we saw in our experimental cybernetics a few slides back we can think about feedback and we talked about this relationship between past present and future a few slides back and we also talked about maybe some of these types of behaviors that are non-random but maybe structured in a certain way and we can think about feedback just sort to revisit the idea of feedback that rosen blue F all gave us which is that feedback was purposeful so we can ask the question is feedback purposeful or simply non- random and so we can think of past present and future relationships as a trip around the feedback Bove

namely that we can take our input we can take the present and have a feedback loop that gives us past information into a new present when we get a new time point and that ends up giving us a future that can then be recycled so this has implications for prediction we can go past present future to prediction or we can have inputs that produce these sort of differently ordered past present futures and use that as the output so feedback mechanisms reorder processes systems with a lot of feedback aren't teleology per se maybe they produce outputs that look teleological but they're not teleological and so going back to the different orders of prediction which were kind of like the final output of our Behavioral um typology what are the different orders of prediction well we have our first second third and fourth or higher order feedback so we can have these feedbacks that have PR past present future but we can nest them together so that we can have a past of a past of a past contributing to the present or to the future and so that's really interesting because what it's suggesting is these higher level feedbacks maybe AR also not teleological but there's some sort of memory or maybe there's some sort of um you know interval and the way it's structured so there's some structural aspect of feedback that might be really interesting that isn't teleological per se but it gives some of these other examples might have some structural significance so if we build a cybernetic model that is really kind of complex and has a lot of interlocked feedbacks and things like that we can get some really interesting results so our different types of higher order feedback you know we can kind of think of these in terms of the derivatives of position so the derivatives of position revisiting that from undergraduate physics you have velocity acceleration jerk snap crackle pop and these are just where you start with velocity and you get a derivative of velocity which is acceleration you get a derivative of acceleration which is jerk and so on and these can be thought of in this way of having these higher order feedbacks or these reordering is a past present future to give you these really complex dynamics in cases of derivatives of position and feedback feedback if it goes farther into the past the effects persist farther into the future so if we go back a couple of feedback loops nested on top of one another and we introduce that to the new present or the new future those effects persist further into the future so we get these really long-term effects in a dynamical system instead of just being local how does this translate into the neuro correlates of behavior well we go back to our idea about behavior sort of being natural teleological and we have to think about like kind of how we kind of assume that behavior does magic things so if we get a behavior like this with higher order feedbacks it tends to resemble the sort of magical aspect it could be something like emergence it could be something like burstiness or something like a set of avalanches that occur and so these are

things that you know you know give us really interesting Dynamics but are no more sort of magical orological than any other kind of behavior and so more fully connect systems are perspective or might be sentient or conscious that's sort of the idea of does magic things so you know we don't want to us Mis attribute sentience or Consciousness to something that isn't and of course this is something that we've have a problem with with modern AI systems that people think they're sentient conscious when in fact they just have this higher order feedback and this sort of um you know behavior that isn't necessarily what people think it is magic things or you know what we talk about like complexity results from embodied function a good example of this is kinematic chains which we'll talk about in a little bit but now we introduced this concept of convolutional Cy cybernetic architectures or ccas and so we go back to our behavioral typology and we ask the question can we better understand the role of purpose or meaning and feedback or temporal order so here we have active Behavior purposeful Behavior theological Behavior and the non- classes of those are nonpurposeful the random behavior and non-feedback or non phological Behavior respectively but we're going to reorder this behavioral classification where now we have non- prediction leading to the order of complexity rather than prediction leading to the order of prediction so we start off with behavior that's active instead of passive then we have nonpurposeful or random Behavior instead of purposeful and non nonactive or passive and purpose for know non- classes then we have feedback which is non-teleological versus feedback which is theological theological feedback is now the non-class non-teleological feedback is now the path to the order of complexity and then instead of prediction being extrapolation we'll have non- prediction which is also extrapolation and then we have our order of complexity first second and third order so the more feedback we have the more complexity we have so this is an example of a convolutional cybernetic architecture we basically have this toy example where we have an input and an output the input is 0000 0 the output is 111 and we have these boxes which process the present to the Future and then our feedbacks uh give us like some past date and it feeds it back to the future or the present so we have a ubiquitous feedback here we have some feed forward elements we have elements of feedback that do certain things to the sequence and the question we ask here is what is the maximal number of steps needed to transform the sequence from 0 0 0 to one11 so we start with z00 Z we put that in we kind of move to our feedback loop our feedback loop does certain things to the sequence and then we end up with 111 and we actually want to know what the maximal number of steps is because we want to know you know how far we can go through these different uh feedback loops and get our

answer this can also be restated in terms of temporal phases past present and future so we also have more explicitly this input of the present we have the future as an output and then we have these different uh Loops that give us the P give us past information and again we're interested in getting the most diverse information possible and so this is again this maximal Loop that we're looking for so we're trying to get the most variety out of these uh convolutional cybernetic architectures instead of trying to find the most efficient path or the tightest path or the most go directed path we can further understand this or restate this in terms of semantic components so we can talk about this in terms of symbols reference and understanding symbol being the input reference being the feedback or part of the feedback loop and then understanding being the output and so we want to accumulate as many reference as possible for a symbol so that we get an output of understanding and again it's the variety that counts not optimizing the process or trying to find a goal directed aspect to this so if we think about how these ccas are assembled uh we can look at the different components of feedback and then we can start to assemble these more complex structures so let's look at first order feedback so CCA has input and in this case we're looking at past present future and maybe alternative Futures and things like that and they all produce a predictive output so first order feedback would be an input going to a future going to the present and naturally the future is just like the present plus the past the present comes in as in it's passed to this feedback which is the past the present and past are added to the Future and then the output is some sort of predictive input so again we have this reordering of past present future in the first order feedback if we have a feed forward only case we have input which leads the present which leads to some complex output but of course as a feed forward only system we just only get the benefit of having the present we don't get to talk about the past in the future it's just like every input results in an output and that output then is complex output within any context then if we put together a larger set of feedback loops we can see this sort of convolved mixed feedback where we have first order feedback second order feedback and a feed forward component so these are all different components producing two different complex outputs so we have an input that gives us the present we get a first order feedback that gives us some past and we add those together to a future state that produces complex output we can also have a feed forward part of that where the input goes to the present and then the present isn't part of that state and then we might have a a second order feedback where we have an input which is the present it passes a first order feed Loop uh feedback loop to the past and another first order or second order feedback grp to the past and those can both be combined into the present which is then

informing a future state which involves a longer memory which has more information and that gives us another complex output but we also have this path from the second order feedback up to an alternative future which then gets passed out as a complex output so you see what we have here is we have an input it goes to the present we sub reference the past we sub reference the past again and that gives us an alternative future so this happens for every cycle so you end up with these three different types of complex output and given this topology we can end up with these different types of characterizations of the system remember again with this convolution what we're introducing is this variety which is actually a concept from uh the ever good regulator theorem and a lot of the work of W. Ashby so we're dealing with producing the maximum amount of variety so we you know we need variety to control systems so this is actually overall the appearance of purposes replaced with an ensemble of positive and negative feedback elements with diffuse effects so remember that we need to have not only a single closed loop feedback but we need to have multiple feedbacks and maybe some open loop feedbacks in there so we can have uh maximize a variety so the imperatives of such a network are to gain processes or to maximize the number of processes and this leads us to maximizing variety and of course variety was a concept that was introduced by W. R. Ashby in the ever good regulator theorem and he talked about how variety is necessary for control but it also creates the conditions for adaptability modularity and modifiability so we have systems that are both not purposeful but they also have this sort of imperative for control but also for other types of conditions in a system that allow it to adapt be modular and be modifiable and so we can also add in polarity we can have different types of effects resulting from feedback as we talked about if we reorder past present and future that's all well and good but this also has relevance to different types of processes that result in systems so in complex systems we have these different effects like Avalanches runaway feedbacks and dampened feedback these are non-linearities and we saw an example of avalanches earlier in our experimental cybernetics and that can result from a number of feedbacks of different polarity and this can produce unpredictable instability we can have things like runaway feedbacks which are basically just the product of positive feedback and that can lead to mass and stability and we can also have something like dampened feedback which is a negative overly negative feedback which sometimes makes things more controllable and so we can have these effects that result from these processes that have a polarity that lead to these different complex systems phenomena that then lead to things that are maybe look they look to logical or purposeful but they're actually just the products of complexity and so if we kind of go back

and look at like how we can model things like connectionist learning and reinforcement learning with cybernetics we can actually take the equations of connectionism and reinforcement learning and put them into this sort of cybernetic system system so this is where you know we want to take modern learning systems or artificial learning systems and understand them using cybernetic approaches and maybe apply them into these kind of ccas and so this this is congruent with hopefield and Sutton's alternative Viewpoint which is that the weights of a network are reinforced by the activity and selection of previous inputs so this is where we have we're looking at Q learning here we're looking at connectionist learning and we're looking at sort of the ways equations are structured and we're thinking about how you know we can get learning from these systems of equations and you know this is a dynamical system you know as much as a learning system so we want to understand kind of how this is how this works and so we started a random set of weights and lack structure due to selection we started a random set of weights and lack of structure due to selection we proceed to a structured set of weights and optimal policies unwittingly we kind of move from a markovian system to a non-markovian deterministic system and so there's a tri question here which is how do we go from something that does not resemble purpose to something that is purposeful the answer is is that purposes a post talk determination so we can model this as habits with weights so we have this input we have these outputs and we have this uh CCA that's that has polarity and we can have habits that are the serve as the inputs and the weights are the polarity and we have non-ic behavior in a directed graph so this directed graph kind of moves towards the end points the outputs and so the lesson here is that purpose is basically whatever we determine these outputs to be so when we observe a behavior a behavior is generated by the CCA graphs we interpret the output the way we'd like to interpret it and as I said before we can have these different outputs that are that look purposeful or they look peely logical but that's just really a post talk determination and we move to wier's Alternative Viewpoint so Wier is more of a cybernetics guy he also has an alternative Viewpoint and his Viewpoint focuses on communication and information which have both a control layer and a semantic layer so again we have to go back to the past present and future formulation so instead of thinking about this in terms of polarity we want to think about this in terms of not just the control but the semantics as well and so we have this past present and future we have an input which is the past we have a future which is the present with past feedback we have the present so the present goes to the Past goes to the future but the future also moves to a present where there's a present with past and future components and then an output so if we run this in time if

we run this as a dynamical system we end up with this the present being sort of you know it's not really additive but it's basically concatenating things onto the present with information from the past and you're informing a future trajectory and you're you can actually generate alternate futures of a system from this and so this there's a control layer a semantic layer and this gives us information about the system both in terms of contingencies and synergies so connectivity and ordering provides both contingencies which is that nodes depend on other nodes and synergies for nodes working together so in this model we can't knock out any one node and have the system work we can't knock out the the feedback and have this sort of Rich self-reference we can only have you know a feedforward model and we've seen that that's much less interesting we also can't knock out the future and we can't knock out the present we have to connect continue with all of our nodes but also these nodes working together produce the synergistic trajectory that gives us rich information about time it gives us this present it gives us a future but it also gives these intermediate states where we get past feedback and we get integration of those this stands in contrast to the imperative of purpose and heliology is cybernetic control and behavior or purposeful or stochastic activity tempered by recursively recombined temporal ordering and so this is again one of the ccas and a CCA is also a series of steps that add recursive functionality so sometimes if we think about this in terms of our internal model memory this is memory with sub referential power so what allows us to have a memory again where we can reference the past of a past of a past and it informs the future of the present and adding present into future and things like that this creates a hierarchical system and high degrees of intentional order so again we can add these things together or we can run these ccas and we can produce this memory with sub referential power and the output can look very organized and hierarchical but the thing that produced that output is not necessarily ordered and it's not necessarily uh there's no overarching goal or any other sort of information going into it so once again thinking about habit and purposeful Behavior we have this paper creatures of habit the Neuroscience of habit and purposeful Behavior this was published in biological Psychiatry so this is where they talk about animal behaviors being composed of Habitual and goal directed categories habitual is non-teleological goal directed is teleological they provide this sort of breakdown where you have teleological versus non-teleological behaviors and deterministic versus stochastic Behavior if we think about teleological and deterministic behaviors we get goal directed behaviors those are things like reaching for a Target or moving towards a goal but of course we know that some of these types of goal-seeking behaviors are chemo like chemotaxis or phototaxis are stochastic and so if we have

teleological and stochasticity we might have something like chemotaxis or phototaxis we have something it's non-teleological and deterministic we have habitual work so we have something that's habitual it's not moving towards a goal but it's deterministic something like uh scratching or like a reflex you have nonic on stochastic behavior however we know that that looks something like a random walk so we can actually characterize these different behaviors in this way the problem with chemotaxis and I put a question Mark in the table so chemotaxis for example which is a movement along chemical gradient we already established that that was not teleological the goal directedness in that case is due to the environment not the internal model and so we can look to Gibsonian information or we can look towards sort of thinking about why it's not teleological and so but it doesn't also doesn't resemble a random walk so that provides us with a problem which is where does this fit into this typology so a clue to kind of where chemotaxis and phototaxis fits into that typology comes from this paper brain like neurodynamics for Behavioral control develop the reinforcement learning and this is from the bioarchive from this year total at all test two scenarios supervised or extent and reinforcement or developmental learning so we have extent and developmental learning only reinforcement learning produces goal directed behavior and neuroactivity similar to biology it provides robust short-term behavioral adaptations to perturbations and it produces an emergent emergence of policies that are not purposeful in and of themselves but lead to things that look the part so this is reminiscent of our CCAs it's also reminiscent of our to our discussion of chemotaxis and phototaxis building a CCA from a small Motif is a form of purposelessness or purposelessness reinforcement learning build up you build up to function and sometimes useless structures and that's okay another example of nonpurposeful behavior that kind of fits into our CCA model comes from feudal substrate Cycles in Biochemistry and so this is the paper stochastic amplification and signaling enzymatic feudal Cycles their noise induced by stability with oscillations this is from pnas so this is where we describe these feudal Cycles is these feedback loops where you have an input of energy there's entic reaction and you get nothing that comes out except entropy or heat so you basically don't have an output your output is entropy it's heat and of course this is a problem because we expect there to be an output we expect this reaction to actually do something and in fact it's creating heat but it seems like that's a lot of work for just heat so we have a biochemical pathway which is a Clos feed back it does thermodynamic work for no reason heat is produced from an energetic input but we don't get any biochemical products and this seems like it's a waste of energy that has no purpose but it's actually important for regulation and other processes it produces

complex higherorder Dynamics or amplification effects throughout a greater biochemical system so if we go from convolution to order we can see that the evolution of cybernetic control is a causal recursive chain of contingencies and so here we look at the contingencies in a CCA we go from one to two to three and then that's a feed forward system and then we can go from 1 to two to three prime to four which feeds back to two and gives us this output four prime or we can go from 1 to two to 3 to four or 1 to two to three prime to four to 5 to 6 to 7 which is an output so we have these different outputs from these different numbered boxes and we have these uh resulting processes so let's take three prime as an example three prime is dependent on two and one and once feedback is operational four and five so four and five actually do feedback to three prime which of course is a dependency so if you knock that dependency out you lose your entire functionality of that part of the network seven on the other hand which is another output at the top is dependent upon 6 5 4 3 Prime 2 and one there's no feedback from delay so you can see the different types of order and how there's a lot of recursive chains that lead to contingencies so again if we think about our feudal cycle our feudal cycle is like something at the bottom I her or feedack back is like something at the top of this diagram and our alternate outputs alternate outputs are at the right and they are of course maximizing for variety and so if we think about the evolution of cybernetic control as a causal recursive chain of contingencies we can think about the kinematic chain which is where we have these dependencies so D is affected by the activities of ab and C we get this chain it's this is this robotic arm that extends out across two joints and the activity in D with kinematic is constrained by A and C and so we can describe it in that way if we go back to these kinds of contingencies we know that seven is constrained by its intermediates and thus it is with the kinematic chain in in the kinematic chain we can do this in a linear fashion we can add on conditionals as convoluted cyber as the convoluted cybernetic Network evolves this restricts the system to a determin istic path behaviorally this leads to interesting consequences for developing sensory motor Loop and a kinematic chain is basically sensory motor physics with contingency so here are examples of a kinematic chain in robotics and we can think about uh embodied robotics here where we have these control systems for these cmatic chains maybe the CCA where we have different types of morphologies and we can design them appropriately we can design and we can design them appropriately we can design behaviors that aren't purposeful aren't te logical but still give us behaviors that we can that look intelligent we can also think about purposeful behavior and fore cognition so our approach to purposeful Behavior can be applied to Fore cognition so we have these different types of

cognition some of them being embodied some of them being embedded some of them being extended which means outside the brain and some of them being inactive if we put this in the framework of Rosenbluth's behavioral classification does for us to rethink that behavioral classification as we did in a number of instances in this talk now there's one insight of 4E that's interesting to us which is that the morphology environment of a system is more important than its internal model so we've been talking about internal models and how these ccas model the internal state of behavior and how that shows that in a lot of cases Theology and purpose are sort of elery but if morphology and environment are more important than the internal model then maybe we need to incorporate our ccas uh or or recast our ccas in terms of morphology and environment as well as an internal state in fact our CCA do that to some extent but it's uh sort of a way to put this behavioral classification on its head and so in the case of 4E purpose is not ascribed to a mind feedback is diffuse and complex prediction its highest orders are not theological they must be open-ended we talk about this in a paper from 2023 and Royal Society interface Focus so what is behavior and what are its properties so yeah we we're kind of revisiting what Rosen Bluth talked about we're looking at a number of examples from neuroscience and from psychology and from other areas animal behavior Robotics and we kind of break down kind of what we're looking at here so Behavior can be intentionality it can be goal directed and supervised and so these are things like control systems par cognition and of course some of those are go directed some of those are regulated complexity then we have Rich higher order complex Dynamics and those are things like Bey reactions which we talked about an embryogenesis which we didn't really talk about but also qualifies in that category so these are D outputs of dynamical systems we also have connectivity and convolution which are things like connectionism avalanches and open-ended learning and then finally we have behaviors that are hardwired but recursive so with our ccas we were modeling some Hardware behaviors in a recursive fashion we also talked about this in terms of non-markovian systems and habitual Behavior we want to go beyond the standard of te logical and purposeful Behavior so a standard way of thinking about behavioral regulation is that inputs are deterministic and outputs or Collective output should be close-ended so if we expand the basic Lo Loop feedback model we get an interconnected network of retrocausal ordering so this is something that we're seeing in the CCA where we talk about the order of causality past present future we have this basic loand feedback model but if we expand that out we get much richer retrocausal ordering which leads to of course to variety but it also leads to these interesting paradoxes and interesting effects of higher order feedback

behavior is often open-ended and this can be characterized by embodied Dynamics unpredictable which can be characterized by a chaotic Dynamics nonnormal which can be characterized by non-markovian behaviors and systems Avalanches of different types power law Dynamics and so forth and variational which means that it's characterized by multiple goals and redundancies and so this requires a reassessment Rosen blue theal typology and has wider consequences for the practice of active inference thank you for your attention uh one of our co-authors Jesse parent uh produced a lengthy blog post on the Rosen blue that all paper talks about its enduring insights and its con uh its context in cybernetics and then we have two papers we talked about in the talk from our group that stasis machines and a primer on Gibson need information so I thank you for your attention e for